

A Systematic Review of Federated Learning for Human Activity Recognition Across Different Data Modalities

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Abstract—Federated Learning (FL) has emerged as a privacy-preserving alternative to centralized machine learning for Human Activity Recognition (HAR). Recent studies have investigated FL across wearable sensor-based, image-based, and multimodal HAR applications. However, differences in data modalities, federated optimization algorithms, communication strategies, and model architectures make it difficult to identify suitable approaches for specific application scenarios. This systematic literature review analyzes recent FL-based HAR studies published between 2024 and 2026. The selected studies are compared based on data modality, datasets, federated algorithms, addressed challenges, and reported performance. Finally, current research gaps and future research directions are identified to guide future developments in privacy-preserving HAR.

Index Terms—Federated Learning, Human Activity Recognition, Wearable Sensors, Multimodal Learning, Privacy Preservation, Differential Privacy

I. INTRODUCTION

Human Activity Recognition (HAR) aims to automatically identify and classify human activities using data collected from wearable sensors, smartphones, cameras, and Internet of Things (IoT) devices [1]. With the widespread adoption of smartphones, wearable devices, and Internet of Things (IoT) technologies, sensor-based HAR has gained significant attention in recent years [2]. However, conventional HAR systems are typically trained using centralized learning, where data are uploaded to cloud servers. This approach raises concerns regarding user privacy, communication overhead, and compliance with data protection regulations, particularly because HAR data often contain sensitive behavioral and health-related information [3].

Federated Learning (FL) addresses these limitations by enabling multiple client devices to collaboratively train a shared model without exchanging raw data. Instead, each client trains the model locally and only model updates are transmitted to a central server, where they are aggregated, commonly using the Federated Averaging (FedAvg) algorithm, to produce an improved global model [4]. Because of its privacy-preserving nature, FL has become a promising paradigm for HAR applications deployed on smartphones, wearable devices, and edge computing platforms.

Recent FL-HAR research has expanded beyond wearable sensor data to include image-based and multimodal activity recognition. These approaches differ considerably in their sensing modalities, datasets, learning algorithms, communication strategies, and privacy mechanisms. Although existing surveys discuss federated learning or HAR separately, few provide a comprehensive comparison of how different data modalities influence the choice of federated learning techniques and the challenges they address. In particular, issues such as communication efficiency, non-IID data, multimodal fusion, and personalization have not been systematically compared across sensing modalities.

To address this gap, this paper presents a systematic literature review of recent FL-HAR research, with a particular focus on comparing federated learning techniques across sensor-based, image-based, and multimodal HAR systems. The review analyzes recent studies according to their application domains, datasets, federated learning algorithms, addressed challenges, and modality-specific characteristics. The study is guided by the following research questions:

- RQ1:** What are the publication trends of FL-based HAR research?
- RQ2:** Which HAR tasks, datasets, and data modalities are most commonly used?
- RQ3:** Which federated learning algorithms and model architectures are employed, and what challenges do they address?

II. BACKGROUND

A. Human Activity Recognition

Human Activity Recognition (HAR) is a field at the intersection of artificial intelligence, pattern recognition, and ubiquitous computing that aims to automatically detect, classify, and monitor a person's physical activities and movements using data collected from various sensors [1], [5]. HAR has numerous applications, including healthcare monitoring, elderly care, rehabilitation, fitness tracking, and smart homes. However, HAR systems face challenges such as data privacy, non-IID data distributions, and limited computational resources on edge devices [6], [7].

1) *Wearable Sensor-Based HAR*: Wearable sensor-based HAR relies on smartphones and wearable devices equipped with sensors such as accelerometers and gyroscopes to recognize daily activities [2]. These devices are still the most common platform for HAR applications due to their low computational cost and wide availability.

2) *Image-Based HAR*: Image-based HAR utilizes RGB, depth, or infrared cameras to recognize human activities from visual information. Compared to wearable sensing, it provides richer contextual information but usually demands more computational and communication resources [1].

3) *Multimodal HAR*: Multimodal HAR utilizes multiple sensing modes of data, such as wearable sensors, images, audio and depth, to increase accuracy and robustness of the recognition process. By combining multiple sources of data, multimodal approach overcomes a lot of limitations of unimodal HAR, but introduces its own difficulties related to data fusion and complexity of computations [8].

B. Federated Learning

Federated Learning (FL) is the distributed machine learning approach that allows multiple clients to train a joint model without direct sharing of their data [9]. Every client trains the model locally on his data and sends the results (model parameters) to the central server, where all the models are aggregated by the specific algorithm (usually FedAvg) to get a better global model [4]. This decentralized framework enhances privacy while reducing the need for centralized data collection, making FL particularly suitable for HAR applications.

The most widely used federated learning optimization algorithm is Federated Averaging (FedAvg), proposed by McMahan *et al.* [10]. The global optimization objective is defined as

$$\min_{\theta} F(\theta) \triangleq \sum_{i=1}^m \frac{n_i}{N} F_i(\theta), \quad (1)$$

where m denotes the number of participating clients, D_i is the local dataset of client i , $n_i = |D_i|$ represents the number of local training samples, and

$$N = \sum_{i=1}^m n_i \quad (2)$$

is the total number of samples across all clients. Here, $F_i(\theta)$ denotes the local objective (loss) function computed using the data of client i . During each communication round, clients perform local model updates and transmit only the model parameters to the server, which aggregates them using FedAvg to obtain an updated global model [11].

III. METHODOLOGY

A. Review Protocol

This systematic literature review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [12] to ensure a transparent and reproducible study selection process.

B. Literature Search Strategy

Studies published between 2024 and 2026 were retrieved from five major academic databases:

- IEEE Xplore
- ACM Digital Library
- arXiv
- MDPI Sensors
- Google Scholar

The primary search query was

("Federated Learning" OR "FL") AND ("Human Activity Recognition" OR "HAR" OR "Activity Recognition")

To ensure coverage of different HAR modalities, additional search queries were conducted.

Wearable Sensor-Based HAR

("Federated Learning") AND ("Human Activity Recognition" OR "HAR") AND (sensor OR wearable OR accelerometer OR gyroscope OR IMU)

Image-Based HAR

("Federated Learning") AND ("Human Activity Recognition" OR "HAR") AND (image OR vision OR RGB OR camera)

Multimodal HAR

("Federated Learning") AND ("Human Activity Recognition" OR "HAR") AND (multimodal OR sensor fusion OR RGB-D OR skeleton)

Retrieved publications were imported into a reference management tool, duplicate records were removed, and the remaining studies were screened based on titles, abstracts, and full-text eligibility according to predefined inclusion and exclusion criteria.

IV. RESULTS

A. Publication trends of FL-based HAR research (RQ1)

Although Federated Learning (FL) was introduced in 2017, this review focuses exclusively on studies published between 2024 and 2025. The selected studies included in this review comprise seven papers published in 2024 and eight papers published in 2025. Quality assessment was performed based on methodology clarity, dataset availability, experimental validation, and reporting completeness.

B. Problem Definition and Data Characteristics (RQ2)

1) *RQ2.1: What HAR Tasks Are Addressed?*: The reviewed studies demonstrate that federated learning (FL) has been applied to a wide range of Human Activity Recognition (HAR) tasks, highlighting its adaptability across diverse application domains. Apart from human activity recognition, many studies focus on addressing practical challenges such as privacy preservation, communication efficiency, personalization, missing modalities, noisy labels, and heterogeneous client data.

Sensor-based HAR remains the dominant research direction. Studies such as EnFed by Mukherjee and Buyya [5],

the framework proposed by Alharbery and Jamil [9], Multi-Frequency FL by Fenoglio et al. [13], LN-FHAR by Sun et al. [14], and the personalized FL approach by Haseeb et al. [7] focus on wearable or smartphone sensing for recognizing daily activities while improving energy efficiency, robustness, scalability, or personalization. These approaches are well suited for smartphones and wearable IoT devices because inertial sensors provide continuous activity monitoring with lower privacy concerns than vision-based systems [2], [3].

Image-based HAR forms another important research category. FedDHAD [15], FedClust [11], and CDFL [16] investigate federated image recognition under heterogeneous data distributions by improving client clustering, adaptive optimization, and communication efficiency. These studies primarily evaluate FL algorithms using standard image datasets to address data heterogeneity.

Recent work shows a growing shift toward multimodal HAR, where multiple sensing modalities are fused to improve recognition performance. MFCPL [17], GraMFedDHAR [18], Rahman and Nguyen’s personalized multimodal FL framework [19], and MHARFedLLM [1] integrate modalities such as inertial sensors, RGB images, audio, depth, and text while addressing challenges including missing modalities, differential privacy, and personalized learning. This trend aligns with recent surveys showing that multimodal sensing provides richer contextual information than unimodal systems [8].

The discussed literature sources provide evidence of the development of FL-HAR in special application areas. In particular, Albogamy [6] implements FL in IoT environments for the purposes of IoMT and healthcare applications, Ndeko et al. [4] use federated split learning combined with differential privacy for privacy-preserving HAR, and Hegiste et al. [20] implement federated action recognition in robotics and smart worker assistance. All in all, the chosen papers suggest that although sensor-based HAR is still the most popular application because of the broad availability of wearable sensors, there is a trend towards multimodal, image-based, healthcare, and privacy-aware HAR [21].

2) *RQ2.2: What Datasets Were Used?*: The selected studies employed a diverse collection of public and custom datasets spanning sensor-based, image-based, and multimodal human activity recognition tasks (Table I). The widespread use of publicly available benchmark datasets highlights the importance of standardized evaluation protocols, enabling fair comparison of federated learning algorithms across studies [21].

Sensor-based datasets were the most frequently used. The UCI-HAR dataset appeared in multiple studies, including Albogamy [6], Ndeko *et al.* [4], Le *et al.* [17], and Rahman and Nguyen [19]. Some other popular datasets included WISDM, MobiAct, PAMAP2, Opportunity, and MEx as the use of smartphones and wearable inertial sensors is popular in activity recognition applications [2], [3].

Image-based studies primarily relied on established computer vision benchmarks, including CIFAR-10, CIFAR-100, SVHN, and TinyImageNet, to evaluate federated learning under non-IID settings [11], [15]. In contrast, CDFL made

use of more representative HAR datasets such as Stanford40, PPMI, and VOC2012 containing realistic human action images [16].

Multimodal datasets were used in fewer number of works, however, gained increased popularity recently. For example, MEx was employed in FedTime-MAGNET [1] and GraMFedDHAR [18], while MFCPL worked with MELD, CMU-MOSEI, and Hateful Memes to evaluate federated learning with heterogeneous and missing modalities [17]. These datasets combine visual, textual, audio, and sensor information, supporting more robust multimodal federated learning [8].

In general, most of the studies used publicly available benchmark datasets ensuring reproducibility and fair comparisons. However, the literature highlights the need for larger multimodal HAR datasets collected under realistic federated settings involving heterogeneous users, wearable devices, smartphones, cameras, and IoT platforms [21].

C. Federated Learning Algorithm and Challenges (RQ3)

1) *RQ3.1: Which Federated Learning Algorithms Were Considered?*: FedAvg remains the dominant optimization algorithm among the reviewed studies owing to its simplicity, communication efficiency, and ease of implementation. Most proposed methods extend the standard FedAvg framework by incorporating personalization strategies, adaptive aggregation, clustering, prototype learning, graph neural networks, or contrastive learning to improve robustness under heterogeneous client environments.

2) *RQ3.2: Which Challenges Were Addressed?*: Table II summarizes the major challenges addressed by each reviewed study.

Among the challenges, non-IID data remains the most significant challenge because user activity patterns naturally vary across individuals, resulting in heterogeneous local data

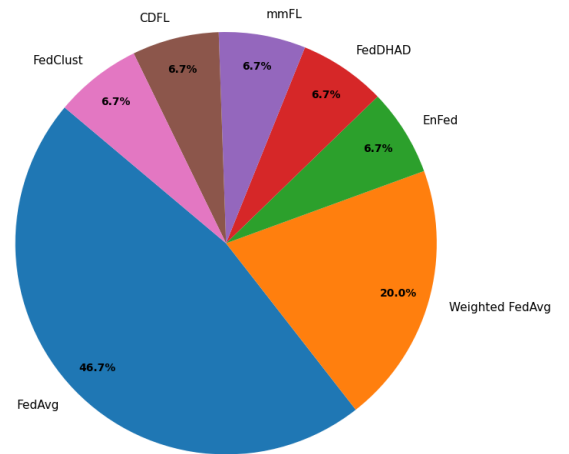


Fig. 1. Distribution of Federated Learning Algorithm

TABLE I
DATASETS USED IN THE REVIEWED STUDIES

Dataset	Type	Classes	Modality	Ref.
Physical Activity & Calorie Burn	Sensor	5 categories	Physical activity sensors	[5]
HARSense	Sensor	6 activities	Accelerometer, Gyroscope	[5]
Human Activity Recognition Database	Sensor	6 activities	Accelerometer, Gyroscope	[5]
CIFAR-10	Image	10 classes	RGB images	[11], [15]
CIFAR-100	Image	100 classes	RGB images	[11], [15]
SVHN	Image	10 classes	RGB images	[11], [15]
TinyImageNet	Image	200 classes	RGB images	[15]
MEx	Multimodal	7 physiotherapy exercises	Accelerometers, Pressure Mat, Depth Camera	[1], [18]
Custom Elderly Dataset	Sensor	6 activities	Accelerometer, Gyroscope, Magnetometer, Barometer	[9]
UCI-HAR	Sensor	6 activities	Smartphone Accelerometer + Gyroscope	[4], [6], [17], [19]
HARTH	Sensor	12 activities	Tri-axial Activity AX3	[6]
HAR70+	Sensor	8 activities	Tri-axial Activity AX3	[6]
Fashion-MNIST	Image	10 classes	RGB images	[11]
WARD	Sensor	18 activities	Accelerometer, Gyroscope	[7]
Hateful Memes	Multimodal	Binary classification	Image + Text	[17]
MELD	Multimodal	4 emotions	Audio + Text + Video	[17]
Stanford40	Image	40 actions	RGB images	[16]
PPMI	Image	Musical activities	RGB images	[16]
VOC2012	Image	Human action recognition	RGB images	[16]
USI-HEAR	Sensor	7 activities	Accelerometer, Gyroscope	[13]
OCOSense	Sensor	7 activities	Accelerometer, Gyroscope, Magnetometer, Pressure	[13]
PAMAP2	Sensor	18 activities	IMUs + Heart Rate Monitor	[14]
OPPORTUNITY	Sensor	17 activities	IMUs, Accelerometers, Localizers	[14]
Custom Skeletal Dataset	Image	8 gestures	RGB-D Video	[20]
CMU-MOSEI	Multimodal	6 emotions	Text + Audio + Video	[19]

TABLE II
CHALLENGES ADDRESSED BY REVIEWED FL-HAR STUDIES

Paper	Non-IID Data	Privacy Method	Comm. Cost	Training Time	Class Imbalance	Multimodal Fusion	Model Compression	Personalization
EnFed [5]	✓	✗	✓	✓	✗	✗	✗	✓
FedDHAD [15]	✓	✗	✓	✓	✗	✗	✓	✓
FedTime-MAGNET [1]	✗	✗	✗	✗	✗	✓	✗	✗
Alharbery [9]	✗	✗	✓	✗	✗	✓	✓	✗
Albogamy [6]	✗	✗	✗	✓	✓	✗	✗	✗
FedClust [11]	✓	✗	✓	✓	✗	✗	✓	✓
FSL + DP [4]	✗	✓	✓	✓	✗	✗	✗	✗
GraMFedDHAR [18]	✓	✓	✗	✗	✗	✓	✗	✗
Haseeb <i>et al.</i> [7]	✗	✗	✓	✗	✗	✗	✓	✓
MFCPL [17]	✗	✗	✓	✗	✗	✓	✗	✓
Hegiste <i>et al.</i> [20]	✗	✗	✓	✓	✓	✗	✓	✗
Rahman & Nguyen [19]	✓	✗	✗	✗	✗	✓	✗	✓
Khazaei <i>et al.</i> [16]	✓	✗	✓	✓	✓	✗	✗	✓
Fenoglio <i>et al.</i> [13]	✗	✗	✗	✗	✗	✗	✗	✗
Sun <i>et al.</i> [14]	✗	✗	✗	✗	✓	✗	✗	✗

distributions that degrade the performance of conventional FedAvg. To address this issue, recent studies have proposed adaptive aggregation, client clustering, prototype learning, graph learning, contrastive learning, and meta-learning [5], [11], [15], [16], [18], [19].

Another major research focus is communication efficiency, as repeated parameter exchanges can significantly increase bandwidth consumption, particularly for image-based and multimodal HAR. Existing studies mitigate this challenge through adaptive dropout, split learning, client selection, pruning, quantization, and prototype-based optimization while maintaining competitive recognition performance [4], [7], [9], [15]–

[17].

Personalized federated learning has also received considerable attention because a single global model often performs poorly for users with different movement patterns and sensing environments. Techniques such as prototype regularization, adaptive aggregation, local fine-tuning, and meta-learning improve user-specific performance while preserving collaborative learning benefits [7], [15], [17], [19]. In parallel, multimodal federated learning has emerged as an important research direction, with graph neural networks, attention mechanisms, encoder–decoder architectures, and prototype learning enabling the integration of heterogeneous data sources, including

wearable sensors, RGB images, audio, and text [1], [17], [18].

Although FL inherently enhances privacy by retaining raw data on client devices, only a limited number of studies incorporated explicit privacy-preserving techniques such as Differential Privacy [4], [18].

Other challenges received comparatively less attention but remain important for practical FL-HAR deployment. Model compression techniques, including pruning and quantization, were adopted to reduce computational and communication overhead on resource-constrained edge devices [7], [9], [11]. Several studies also improved training efficiency through adaptive dropout, early stopping, and communication-efficient optimization [4], [15]. In contrast, challenge such as class imbalance was addressed by only a few studies, indicating that this remains relatively underexplored despite their potential impact on large-scale real-world federated learning systems.

Overall, the reviewed studies demonstrate a clear transition from conventional FedAvg-based learning toward adaptive aggregation, personalization, communication-efficient optimization, and multimodal learning. While non-IID data and communication efficiency remain the dominant research challenges, privacy-preserving techniques and scalable multimodal deployment continue to represent important directions for future FL-HAR research [21].

V. DISCUSSION

The analysis of the selected studies reveals a clear evolution in the application of federated learning (FL) to Human Activity Recognition (HAR). Early studies mainly focused on preserving the privacy of the users while at the same time maintaining the performance of recognition techniques within FedAvg-based frameworks. The more recent studies have been more concerned about the challenges associated with deploying these techniques, such as non-IID data, efficient communication, lack of certain types of modality, noisy labels, and personalized learning.

The reviewed studies also demonstrate a strong reliance on publicly available benchmark datasets, which facilitate standardized evaluation and fair comparison of proposed methods. Only a limited number of studies employed custom datasets, highlighting the need for larger and more diverse benchmark datasets that better represent real-world federated environments.

When compared across different sensing modalities, it becomes obvious that sensor-based HAR is the most mature application area for FL technology. Since inertial sensors produce relatively low-dimensional data, the recognition techniques built using them have high performance but low communication and computational overhead. However, image-based HAR poses more challenges due to the need for larger models for handling high-dimensional images, which leads to higher computation and communication costs. Therefore, recent works have applied adaptive aggregation, client clustering, and contrastive learning to boost the convergence of FL-HAR in cases of heterogeneous distributions of data samples. Multimodal HAR has emerged as the most promising research direction

by combining complementary information from heterogeneous sensing modalities, including wearable sensors, RGB images, depth cameras, audio, and text. To solve the corresponding problems of modality heterogeneity and multimodal fusion, recent studies utilize techniques like prototype learning, graph neural networks, attention mechanism, and transformers.

Despite the significant progress reported in the reviewed studies, several research gaps remain.

First, although numerous approaches have been proposed to mitigate *non-IID data*, their effectiveness has primarily been demonstrated on benchmark datasets with a limited number of clients. Their scalability and robustness in large-scale heterogeneous deployments remain largely unexplored.

Second, *communication efficiency* continues to be a major challenge, particularly for image-based and multimodal HAR. While pruning, quantization, split learning, and adaptive communication reduce transmission costs, few studies evaluate these methods under realistic deployment conditions considering network latency, bandwidth limitations, and energy consumption.

Third, *privacy preservation* remains insufficiently explored. Although federated learning prevents raw data from leaving client devices, exchanged model updates are still vulnerable to inference attacks. Most reviewed studies rely solely on the inherent privacy advantages of FL, while advanced techniques such as Secure Aggregation, Homomorphic Encryption, and Secure Multi-Party Computation have received limited attention.

Finally, *multimodal federated learning* still faces challenges related to missing modalities, heterogeneous feature spaces, and incomplete sensing environments. Although MFCPL addresses missing modalities through prototype learning [17], robust multimodal learning under realistic deployment scenarios remains an open research problem.

A. Future Research Directions

Based on the findings of this review, several promising directions can guide future research on federated learning for Human Activity Recognition (HAR).

Balancing Privacy and Utility: Future FL-HAR frameworks should investigate the integration of advanced privacy-preserving techniques, such as Differential Privacy (DP), Secure Multi-Party Computation (SMPC), and Homomorphic Encryption (HE) [4], [18], [16]. A key challenge will be achieving an effective balance between privacy protection, computational overhead, and recognition performance [4], [18].

Communication-Efficient Federated Learning: Communication efficiency remains a critical requirement for deploying FL-HAR systems on resource-constrained wearable and edge devices. Future research should jointly optimize recognition accuracy, communication overhead, computational complexity, and energy consumption [5], [16].

Scalable Multimodal FL-HAR: Multimodal federated learning is expected to play an increasingly important role in future HAR systems by combining wearable sensors, images,

TABLE III
COMPARISON OF SENSOR-BASED, IMAGE-BASED, AND MULTIMODAL HAR IN FEDERATED LEARNING

Aspect	Sensor-based HAR	Image-based HAR	Multimodal HAR
Privacy	High	Low	Medium
Data Size	Small	Large	Very Large
Communication Cost	Low	High	Medium
Computation	Low	High	Very High
Models	LSTM, MLP, Bi-LSTM	LeNet, CNN, VGG, ResNet	T5, DART-CNN, MAGNET

audio, and textual information. However, effectively managing heterogeneous modalities while maintaining scalability and computational efficiency remains an open challenge [1], [19]. Future work should focus on robust multimodal fusion, missing modality handling, parameter-efficient fine-tuning, and scalable model architectures that support real-world federated deployments [1], [17], [18].

VI. CONCLUSION

This paper presented a systematic literature review of federated learning for Human Activity Recognition (HAR) with a comparative analysis across sensor-based, image-based, and multimodal sensing modalities. By reviewing fifteen recent studies, the paper examined how different modalities influence the selection of federated learning algorithms, datasets, optimization strategies, and the challenges addressed by existing approaches.

The review shows that sensor-based HAR remains the most mature research direction due to its communication efficiency and suitability for wearable devices, while image-based and multimodal HAR increasingly require advanced techniques such as adaptive aggregation, graph learning, prototype learning, clustering, and personalized federated learning to cope with heterogeneous data and richer feature representations. The comparative analysis further demonstrates that each sensing modality presents distinct trade-offs in privacy, computational complexity, communication cost, and model design.

Overall, there is no single federated learning technique suitable for all HAR modalities. Instead, the choice of optimization strategy depends on the characteristics of the sensing modality and the target application. Future research should focus on communication-efficient multimodal federated learning, stronger privacy-preserving mechanisms, realistic multimodal benchmark datasets, and scalable personalization methods to enable robust deployment in real-world HAR applications.

REFERENCES

- [1] A. Bandyopadhyay, R. Basu, T. Sen, and S. Das, "MHARFedLLM: Multimodal human activity recognition using federated large language model," *arXiv preprint arXiv:2508.01701*, Aug. 2025.
- [2] A. Wang, G. Chen, J. Yang, S. Zhao, and C.-Y. Chang, "A comparative study on human activity recognition using inertial sensors in a smart phone," *IEEE Sensors Journal*, vol. 16, no. 11, pp. 4566–4578, Jun. 2016.
- [3] M. T. Khan, E. Ever, S. Eraslan, and Y. Yesilada, "Human activity recognition using binary sensors: A systematic review," *Inf. Fusion*, vol. 115, p. 102731, Mar. 2024.
- [4] J. Ndeko, S. Shaon, A. Beal, A. Sahoo, and D. C. Nguyen, "Federated split learning for human activity recognition with differential privacy," *arXiv preprint arXiv:2411.06263*, Nov. 2024.
- [5] A. Mukherjee and R. Buyya, "EnFed: An energy-aware opportunistic federated learning in resource constrained environments for human activity recognition," *arXiv preprint arXiv:2412.00768*, Dec. 2024.
- [6] F. R. Albogamy, "Federated learning for IoMT-enhanced human activity recognition with hybrid LSTM-GRU networks," *Sensors*, vol. 25, no. 3, p. 907, Feb. 2025.
- [7] A. Haseeb, I. Cleland, C. Nugent, and J. McLaughlin, "Efficient and personalized federated learning for human activity recognition on resource-constrained devices," *Appl. Sci.*, vol. 16, no. 2, p. 700, Jan. 2026.
- [8] M. B. Shaikh, D. Chai, S. M. S. Islam, and N. Akhtar, "From CNNs to transformers in multimodal human action recognition: A survey," *ACM Trans. Multimedia Comput. Commun. Appl.*, vol. 20, no. 8, pp. 1–24, Aug. 2024.
- [9] R. A. Alharberly and F. Jamil, "Federated learning framework for real-time activity and context monitoring using edge devices," *Sensors*, vol. 25, no. 4, p. 1266, Feb. 2025.
- [10] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273–1282.
- [11] M. S. Islam, S. Javaherian, F. Xu, X. Yuan, L. Chen, and N. F. Tzeng, "FedClust: Tackling data heterogeneity in federated learning through weight-driven client clustering," in *Proc. 53rd Int. Conf. Parallel Process. (ICPP)*, Aug. 2024.
- [12] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, and et al., "The PRISMA 2020 statement: An updated guideline for reporting systematic reviews," *BMJ*, vol. 372, p. n71, Mar. 2021.
- [13] D. Fenoglio, M. Li, D. Casnici, M. Laporte, S. Gashi, S. Santini, M. Gjoreski, and M. Langheinrich, "Multi-frequency federated learning for human activity recognition using head-worn sensors," *arXiv preprint*, 2024.
- [14] H. Sun, J. Yao, X. Li, Y. Liu, and H. Gu, "Robust two stages federated learning for sensor based human activity recognition with label noise," *Sci. Rep.*, vol. 15, p. 17227, May 2025.
- [15] J. Liu, B. Ma, Q. Yu, R. Jin, J. Zhou, Y. Zhou, H. Dai, H. Wang, D. Dou, and P. Valduriez, "Efficient federated learning with heterogeneous data and adaptive dropout," *arXiv preprint arXiv:2507.10430*, Jul. 2024.
- [16] E. Khazaei, A. Esmailzadeh, B. Taha, and D. Hatzinakos, "CDL: Efficient federated human activity recognition using contrastive learning and deep clustering," *arXiv preprint arXiv:2407.12287*, Jul. 2024.
- [17] H. Q. Le, C. M. Thwal, Y. Qiao, Y. L. Tun, M. N. H. Nguyen, E. N. Huh, and C. S. Hong, "Multimodal federated cross prototype learning with missing modalities," *arXiv preprint arXiv:2401.13898*, Jan. 2024.
- [18] L. Halder, T. Sen, and S. Palit, "GRAMFEDDHAR: Graph based multimodal differentially private federated HAR," *arXiv preprint arXiv:2509.05671*, Sep. 2025.
- [19] R. Rahman and D. C. Nguyen, "Multimodal federated learning with model personalization," in *2024 Optim. Mach. Learn.*, 2024.
- [20] V. Hegiste, V. Goyal, T. Legler, and M. Ruskowski, "Federated action recognition for smart worker assistance using FastPose," *arXiv preprint arXiv:2508.14113*, Aug. 2025.
- [21] A. Grataloup and M. Kurpicz-Briki, "A systematic survey on the application of federated learning in mental state detection and human activity recognition," *Front. Digit. Health*, vol. 6, p. 1495999, Oct. 2024.